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Studierichting: Informatica

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$$\begin{aligned} \int \frac{1}{\sqrt{x} + \sqrt[3]{x}} dx &\stackrel{x=t^6}{=} 6 \int \frac{t^5}{t^3 + t^2} dt \\ &= 6 \int t^2 dt - 6 \int t dt + 6 \int dt - 6 \int \frac{1}{t+1} dt \\ &= 2t^3 - 3t^2 + 6t - 6 \ln |t+1| + c \\ &\stackrel{t=\sqrt[6]{x}}{=} 2\sqrt{x} - 3\sqrt[3]{x} + 6\sqrt[6]{x} - 6 \ln |\sqrt[6]{x} + 1| + c \end{aligned}$$

References